

Implement a file copy utility using low-level Linux file I/O system calls (open, read, write, lseek, close). Compare its performance with standard library functions (fopen, fread, fwrite).

Code:

```
#include <stdio.h>
#include <fcntl.h>
#include <unistd.h>
#include <stdlib.h>
#include <time.h>

/* -----
   LOW-LEVEL COPY
   ----- */

void low_level_copy(char *source, char *destination)
{
    int src, dest;
    char buffer[1024];
    int bytes;

    src = open(source, O_RDONLY);

    if (src == -1)
    {
        perror("Source file");
        return;
    }

    dest = open(destination, O_WRONLY | O_CREAT | O_TRUNC, 0644);
```

```
if (dest == -1)
{
    perror("Destination file");
    close(src);
    return;
}

/* Show current file position */
printf("Initial position: %ld\n", lseek(src, 0, SEEK_CUR));

while ((bytes = read(src, buffer, sizeof(buffer))) > 0)
{
    write(dest, buffer, bytes);
}

/* Move back to beginning */
lseek(src, 0, SEEK_SET);

printf("Position after lseek: %ld\n",
        lseek(src, 0, SEEK_CUR));

close(src);
close(dest);
}
```

```

/* -----
   STANDARD LIBRARY COPY
   ----- */

void standard_copy(char *source, char *destination)
{
    FILE *src;
    FILE *dest;

    char buffer[1024];
    int bytes;

    src = fopen(source, "rb");

    if (src == NULL)
    {
        perror("Source file");
        return;
    }

    dest = fopen(destination, "wb");

    if (dest == NULL)
    {
        perror("Destination file");
    }
}

```

```

        fclose(src);
        return;
    }

    while ((bytes = fread(buffer, 1, sizeof(buffer), src)) > 0)
    {
        fwrite(buffer, 1, bytes, dest);
    }

    fclose(src);
    fclose(dest);
}

/* -----
   MAIN
   ----- */

int main()
{
    char source[100];
    char low_destination[100];
    char standard_destination[100];

    clock_t start, end;
|

```

```

printf("Enter source file: ");
scanf("%99s", source);

printf("Enter low-level destination file: ");
scanf("%99s", low_destination);

printf("Enter standard-library destination file: ");
scanf("%99s", standard_destination);

/* -----
   LOW-LEVEL COPY
   ----- */

printf("\n===== LOW-LEVEL COPY =====\n");

start = clock();

low_level_copy(source, low_destination);

end = clock();

double low_time =
    (double)(end - start) / CLOCKS_PER_SEC;

```

```

printf("Low-level copy completed.\n");
printf("Time: %f seconds\n", low_time);

/* -----
   STANDARD LIBRARY COPY
   ----- */

printf("\n===== STANDARD LIBRARY COPY =====\n");

start = clock();

standard_copy(source, standard_destination);

end = clock();

double standard_time =
    (double)(end - start) / CLOCKS_PER_SEC;

printf("Standard library copy completed.\n");
printf("Time: %f seconds\n", standard_time);

/* -----
   COMPARISON
   ----- */

```

```

/* -----
   COMPARISON
   ----- */

printf("\n===== COMPARISON =====\n");

printf("Low-level functions      : open, read, write, lseek, close\n");
printf("Standard functions       : fopen, fread, fwrite, fclose\n");

printf("Low-level time           : %f seconds\n",
       low_time);

printf("Standard library time    : %f seconds\n",
       standard_time);

printf("===== \n");

return 0;
}

```

Output:

```
geya@LenovoL480:~/OS/Practical/Week 9$ ./task1
Enter source file: input.txt
Enter low-level destination file: copy_low.txt
Enter standard-library destination file: copy_standard.txt

===== LOW-LEVEL COPY =====
Initial position: 0
Position after lseek: 0
Low-level copy completed.
Time: 0.000276 seconds

===== STANDARD LIBRARY COPY =====
Standard library copy completed.
Time: 0.000109 seconds

===== COMPARISON =====
Low-level functions      : open, read, write, lseek, close
Standard functions      : fopen, fread, fwrite, fclose
Low-level time           : 0.000276 seconds
Standard library time    : 0.000109 seconds
=====
geya@LenovoL480:~/OS/Practical/Week 9$ |
```